SOC THREAT DETECTION & INCIDENT RESPONSE

Syntexel X

Explainable, rule-based detection that turns raw login logs into investigated, report-ready alerts.

Python · Flask · SQLite · MITRE ATT&CK · pytest

Learning prototype · synthetic data only · 01

developed by k.sai kowshiik reddy

THE PROBLEM

Analysts must find the few suspicious eve hiding among thousands of routine ones

Logs everywhere

Every login, access and error creates a log line.

Signal vs noise

One mistyped password and one brute-force attempt look alike in a single line.

Context decide

Patterns across time, and account separate

Syntexel X automates the first pass and keeps the evidence attached, so the analyst can the call.

WHAT IT DOES

Four capabilities, each built and tested

Log analysis

Read, validate and store JSON authentication events.

Threat detection

Rules flag repeated failures, success
failures and combined indicators.

Alert dashboard

Severity, filters, evidence and MITRE ATT&CK context.

Incident tracking

Status, owner and notes per alert, p
downloadable report.

HOW IT WORKS

From raw logs to a reviewed alert

01
Logs
JSON login events

→

02
Parser
Checks fields, timestamps, event types

→

03
Rules
Group failures, correlate successes

→

04
Alerts
Severity, evidence, MITRE; saved in SQLite

→

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inv
ex

Python 3 | Flask | SQLite | Bootstrap 5 + Chart.js | pytest

LOG DATA

Every event is validated before detection

```
{
  "timestamp": "2026-10-09T10:45:00",
  "event_type": "login_failed",
  "username": "admin",
  "source_ip": "198.51.100.23",
  "status": "failure"
}
```

The parser checks

- All five required fields are present
- Timestamp is valid ISO-8601
- Event type is a known value
- Source IP is a valid address

Bad records become a Low data-quality alert ins
silently dropped.

DETECTION RULES

Four simple rules an analyst can explain

Rule	Logic	Severity
Repeated failed logins	5 or more failures from one IP within 5 minutes	High
Success after repeated failures	Same account and IP succeeds after 3 or more failures in 5 minutes	Medium (
Suspicious event combination	The same IP triggers both rules above	Critical
Invalid log record	Required field missing or malformed	Low

WORKED EXAMPLE

One IP, one account, three alerts

10:45 – 10:49	8 failed logins for admin from 198.51.100.23, one every 40 seconds
10:50	Successful login for admin from the same IP
Detection	Repeated Failed Logins (High), Success After Repeated Failures (High), Suspicious Combination (Critical)
Response	Confirm with the account owner, reset credentials, revoke sessions, block the IP,

An alert is an indicator, n proof.

Every finding is a possible match for the analyst to confirm. Each alert starts as New, and a person decides whether it is an incident, a false positive or something to escalate.

SEVERITY AND MITRE ATT&CK

Priority and context for every alert

Critical	Several independent indicators from one IP
High	Burst of failures, or success after 5+ failures
Medium	Success after 3–4 failures
Low	Data-quality issue in the logs

T1110 Brute Force
Repeated failures against one account

T1110.003 Password Spraying
Failures across 3 or more accounts

T1078 Valid Accounts
Success after failures

Mappings are possible matches, not confirmed attacks · 09

THE DASHBOARD

Four pages, one investigation workflow

Dashboard

Alert totals, open investigations, severity chart, hourly trend

Alerts

Filter by severity, source IP, rule, status and date

Alert detail

Evidence, MITRE mapping, impact, next steps, notes

Investiga

Every case wi
owner and lat

Investigation status

New → Investigating → Escalated → Resolved or False Positive

INCIDENT REPORT

Every alert exports as a client-ready repo

1 Finding
What was detected, in plain words

2 Affected asset
Accounts and source IP

3 Evidence
Timestamps, counts, even

4 Severity
Per the lab's defined rules

5 Potential impact
What could happen if malicious

6 Recommended a
Validate, contain, escalate

7 MITRE ATT&CK
Possible technique match

8 Status and notes
Owner and analyst notes

9 Retest
Confirm the pattern stopp

TESTING

Verified, not just built

27
automated tests passing

7
scenarios in the sample logs

6
alerts raised, none fro
normal activity

- Five failures in five minutes alert; four do not
- Failures spread over hours do not alert
- Re-ingesting the same logs creates no duplicates
- Forms without a CSRF token are rejected

Parser, detection rules, services and Flask routes all covered · 12

HONEST SCOPE

Limitations and next steps

Limitations today

- Batch ingestion only, not real time
- No login or roles on the dashboard
- No GeoIP or known-IP context
- Single-node SQLite
- Synthetic data only

Next steps

- Real Windows and Linux auth logs
- Allow lists and GeoIP enrichment
- Dashboard authentication
- MySQL or PostgreSQL
- Email or Slack alerting